

7th Grade Standard Mathematics

WEEK 1 — DAILY LESSON PLANS

Unit 1: Introduction | Lessons 1.1 – 1.3 | Q1 | 2026–2027

Lesson Format (45 min)

□ Bell Work (5 min) → □ Benchmark / Goal / Targets (4 min) → □ Main Idea + Vocabulary (7 min)
 I Do — Teacher Models (6 min) → We Do — Together (7 min) → You Do — Independent (9 min) → □ Exit Slip (5 min)

Day	Lesson	Title	Benchmark(s)
1	1.1 Pt 1	What is a Statistical Question?	MA.6.DP.1.1
2	1.1 Pt 2	Interpreting Histograms & Line Plots	MA.6.DP.1.4
3	1.2 Pt 1	Ratio Relationships	MA.6.AR.3.1, 3.2
4	1.2 Pt 2	Rates, Unit Rates & Percentages	MA.6.AR.3.2, 3.4
5	1.3 Pt 1	Rational Numbers on the Number Line	MA.6.NSO.1.1, 1.2

DAY

1

What is a Statistical Question?

Lesson 1.1 — Part 1 | Benchmark: MA.6.DP.1.1

Unit 1: Introduction — Week 1 — 45 min

BELL WORK

5 min

Directions: Write the question AND your answer in your notebook.

Question: Your teacher asks the class: "How tall are you?" and records everyone's answer. Every student gives a different number. Why might that be?

Answer: Because heights vary — different people have different heights. When answers vary, the question generates data.

BENCHMARK — LEARNING GOAL — LEARNING TARGETS

4 min

Benchmark

MA.6.DP.1.1 — Recognize and formulate a statistical question that would generate numerical data.

Learning Goal

Students will identify, distinguish, and formulate statistical questions.

Target 1

I can tell the difference between a statistical question and a non-statistical question.

Target 2

I can rewrite a non-statistical question to make it statistical.

MAIN IDEA + VOCABULARY

7 min

Main Idea: A **statistical question** is one where we expect the answers to vary — different people or things will give different responses. A **non-statistical question** has only one correct answer.

Term	Definition
Statistical question	A question that expects variability in the answers collected (e.g., "How tall are students in our school?")
Non-statistical question	A question with only one definite answer (e.g., "How tall is the Eiffel Tower?")
Data	A collection of facts, numbers, or information gathered to answer a question
Variability	The differences or spread in a set of data values

I DO — Teacher Models

6 min

Example Question: "How many siblings do students in 7th grade have?"

Step 1: Ask — does this question have ONE answer, or will different people give DIFFERENT answers?

Step 2: Some students have 0 siblings, some have 1, some have 3 or more. The answers VARY.

Step 3: Because answers vary, this IS a statistical question. It generates data we can analyze.

Non-statistical version: "How many siblings does Marcus have?" — only ONE answer. NOT statistical.

WE DO — Together

7 min

Problem: Decide together — is each question statistical or not statistical? If not, rewrite it.

1. "How many days are in January?"

→ One answer (31 days) — NOT statistical. Rewrite: "How many days are in each month of the year?"

2. "How many hours of sleep do 7th graders get each night?"

→ Answers vary per student — STATISTICAL.

Discuss: What's the key signal that a question is statistical? (The answers vary!)

YOU DO — Independent Practice

9 min

Directions: For each question below, write S (statistical) or NS (non-statistical). If NS, rewrite it to make it statistical.

1. What is the most popular sport at your school?
2. What temperature is water when it boils?
3. How long does it take students to get to school?
4. How old is your math teacher?

Practice Book Reference: Lesson 1, Questions 1–4

IXL PRACTICE ASSIGNMENT

Homework / Early Finisher

Assigned Skill: Statistical Questions

Benchmark: MA.6.DP.1.1

URL: <https://www.ixl.com/math/grade-6/statistical-questions>

Directions: Log in to IXL and complete this skill. Aim for a SmartScore of 70 or higher.

Teacher Tip: Check IXL Trouble Spots after class to identify students who need reteaching.

IXL PRACTICE ASSIGNMENT

Homework / Early
Finisher

Assigned Skill: Interpret Histograms and Line Plots

Benchmark: MA.6.DP.1.4

URL: <https://www.ixl.com/math/grade-6/interpret-histograms>

Directions: Log in to IXL and complete this skill. Aim for a SmartScore of 70 or higher.

Teacher Tip: Check IXL Trouble Spots after class to identify students who need reteaching.

IXL PRACTICE ASSIGNMENT

Homework / Early
Finisher

Assigned Skill: Ratios and Rates: Word Problems

Benchmark: MA.6.AR.3.1-3.2

URL: <https://www.ixl.com/math/grade-6/ratios-and-rates-word-problems>

Directions: Log in to IXL and complete this skill. Aim for a SmartScore of 70 or higher.

Teacher Tip: Check IXL Trouble Spots after class to identify students who need reteaching.

IXL PRACTICE ASSIGNMENT

Homework / Early
Finisher

Assigned Skill: Find the Percent: Tax, Discount and More

Benchmark: MA.6.AR.3.4

URL: <https://www.ixl.com/math/grade-6/find-the-percent-of-a-number-tax-discount-and-more>

Directions: Log in to IXL and complete this skill. Aim for a SmartScore of 70 or higher.

Teacher Tip: Check IXL Trouble Spots after class to identify students who need reteaching.

IXL PRACTICE ASSIGNMENTHomework / Early
Finisher**Assigned Skill:** Rational Numbers on Number Lines**Benchmark:** MA.6.NSO.1.1-1.2**URL:** <https://www.ixl.com/math/grade-6/rational-numbers-on-number-lines>**Directions:** Log in to IXL and complete this skill. Aim for a SmartScore of 70 or higher.**Teacher Tip:** Check IXL Trouble Spots after class to identify students who need reteaching.**EXIT SLIP**

5 min

On a separate slip of paper or on the bottom of your notebook page, answer:

1. Write **ONE** statistical question you could ask about students in this classroom.
2. Explain in one sentence **WHY** your question is statistical.

Teacher Note: Collect slips to identify students who need additional support before Day 2.

DAY
2

Interpreting Histograms & Line Plots

Lesson 1.1 — Part 2 | Benchmark: MA.6.DP.1.4

Unit 1: Introduction — Week 1 — 45 min

☐ BELL WORK

5 min

Question: Look at the histogram on the board showing students' test scores. How many students scored between 70–79? What is the range of the data?

Answer: Count the bar height for 70–79. Range = highest score – lowest score. (Teacher fills in specific values when displaying histogram.)

☐ BENCHMARK — LEARNING GOAL — LEARNING TARGETS

4 min

Benchmark

MA.6.DP.1.4 — Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including symmetry, skewness, gaps, clusters, outliers, and range.

Learning Goal

Students will describe and interpret data distributions from histograms and line plots.

Target 1

I can identify and describe features of a data distribution: symmetry, skewness, gaps, clusters, outliers, and range.

Target 2

I can interpret information from a histogram or line plot in a real-world context.

☐ MAIN IDEA + VOCABULARY

7 min

Main Idea: Data doesn't just tell us numbers — it tells us a story about how values are spread out. A distribution can be symmetric, skewed, have clusters, gaps, or outliers.

Term	Definition
Distribution	The way data values are spread across the range (where they fall on a display)
Symmetric	When both sides of a display are mirror images — data balanced around the center
Skewed	When data is bunched toward one side — skewed right = tail stretches right; skewed left = tail stretches left
Cluster	A group of data points that are close together
Gap	An area in the distribution where there are no data values

Outlier	A data value that is much higher or lower than most other values
Range	The difference between the maximum and minimum values in a data set

I DO — Teacher Models

6 min

Teacher displays a histogram of daily high temperatures for one month.

I say: "Let me describe what I see using our vocabulary."

- **Shape:** The data appears skewed right — most days cluster at lower temperatures with a few very hot days pulling the tail right.
- **Cluster:** Most values fall between 65–80 degrees.
- **Outlier:** One bar stands alone near 100 degrees — that's an outlier.
- **Range:** $100 - 60 = 40$ degrees.

WE DO — Together

7 min

Display a line plot of student reading times (minutes per day). Work through together:

1. What is the range of the data?
2. Where do you see a cluster?
3. Are there any outliers? How do you know?
4. Is the data symmetric or skewed? How can you tell?

Guide students to use full sentences: "The data is skewed ____ because ____"

YOU DO — Independent Practice

9 min

Use the histogram on the board showing homework completion times.

Answer the following in your notebook:

1. What is the range of homework times?
2. Describe the shape — is it symmetric or skewed? Explain.
3. Identify any clusters and gaps.
4. Are there any outliers? How do you know?

Practice Book Reference: Lesson 1, Questions 5–10

EXIT SLIP

5 min

Look at the histogram provided. Use at least TWO vocabulary words to describe the distribution.

Sentence starters: "The data is ____ because ____" | "I notice a cluster at ____" | "The range is ____"

Teacher Note: Look for correct use of skewed, cluster, outlier — these are key terms for MA.7.DP.1.1 later in the year.

DAY

3

Ratio Relationships

Lesson 1.2 — Part 1 | Benchmark: MA.6.AR.3.1, MA.6.AR.3.2

Unit 1: Introduction — Week 1 — 45 min

☐ BELL WORK

5 min

Question: A recipe uses 2 cups of sugar for every 3 cups of flour. Write this ratio in three different ways.

Answer: 2:3 | 2 to 3 | $\frac{2}{3}$ (All represent the same ratio of sugar to flour.)

☐ BENCHMARK — LEARNING GOAL — LEARNING TARGETS

4 min

Benchmark

MA.6.AR.3.1 — Given a real-world context, write and interpret ratios to show relative sizes of two quantities. MA.6.AR.3.2 — Apply ratio reasoning to solve mathematical and real-world problems.

Learning Goal

Students will write, interpret, and apply ratio relationships using tables and double number lines.

Target 1

I can write and interpret a ratio relationship in a real-world context.

Target 2

I can use a double number line or ratio table to find equivalent ratios.

☐ MAIN IDEA + VOCABULARY

7 min

Main Idea: A ratio is a way to compare two quantities. We can show equivalent ratios using a double number line or a ratio table — both show how the relationship scales up or down.

Term	Definition
Ratio	A comparison of two quantities — can be written as $a:b$, a to b , or $\frac{a}{b}$
Equivalent ratios	Ratios that represent the same relationship (like equivalent fractions)
Double number line	Two parallel number lines used to model a ratio relationship
Ratio table	A table showing pairs of equivalent ratios
Rate	A ratio comparing two quantities with different units (e.g., miles per hour)
Unit rate	A rate where the second quantity equals 1 (e.g., 60 miles per 1 hour)

I DO — Teacher Models

6 min

Problem: A car travels 120 miles in 2 hours. How far will it travel in 5 hours?

Step 1: Write the ratio: 120 miles : 2 hours → Unit rate = 60 miles per 1 hour

Step 2: Build a ratio table:

Hours	1	2	5
Miles	60	120	300

Answer: The car travels 300 miles in 5 hours.

WE DO — Together

7 min

Problem: The ratio of wins to losses for the Tampa Bay Thunder soccer team is 4:3. If the team won 16 games, how many did they lose?

Work through together:

1. What is the unit for each quantity? (wins and losses)
2. Set up the ratio table with the given ratio 4:3.
3. Scale up to find losses when wins = 16.

Then find: If the team played 35 total games, how many were wins?

YOU DO — Independent Practice

9 min

Problem: Maria earns \$45 for every 3 hours she works.

1. Write the ratio of dollars earned to hours worked.
2. Find the unit rate (dollars per 1 hour).
3. Complete the ratio table: 1 hr, 3 hrs, 5 hrs, 8 hrs.
4. How much will Maria earn in 7 hours?

Practice Book Reference: Lesson 2, Questions 1–4

EXIT SLIP

5 min

The ratio of cats to dogs at a shelter is 3:5.

- A. If there are 15 cats, how many dogs are there? Show your work.
- B. What is the unit rate of dogs per cat?

DAY

4

Rates, Unit Rates & Percentages

Lesson 1.2 — Part 2 | Benchmark: MA.6.AR.3.2, MA.6.AR.3.4

Unit 1: Introduction — Week 1 — 45 min

BELL WORK

5 min

Question: A store sells 5 apples for \$2.50. What is the price per apple (the unit rate)?

Answer: $\$2.50 \div 5 = \0.50 per apple. The unit rate is \$0.50 per 1 apple.

BENCHMARK — LEARNING GOAL — LEARNING TARGETS

4 min

Benchmark

MA.6.AR.3.2 — Apply ratio reasoning to solve mathematical and real-world problems.
MA.6.AR.3.4 — Apply ratio reasoning and use proportions to solve mathematical and real-world percent problems.

Learning Goal

Students will find unit rates and use ratio reasoning to solve real-world percent problems.

Target 1

I can find and interpret unit rates in real-world contexts.

Target 2

I can use ratio reasoning to find percent discounts and sale prices.

MAIN IDEA + VOCABULARY

7 min

Main Idea: A percent is a special ratio that compares a number to 100. We can use ratio tables or proportions to find discounts, taxes, and sale prices in real-world situations.

Term	Definition
Percent	A ratio that compares a quantity to 100; means 'per hundred' (e.g., 25% = 25/100)
Discount	An amount subtracted from the original price
Sale price	The final price after a discount is applied (Original Price – Discount)
Tax	A percentage added to a price (e.g., sales tax)
Proportion	An equation stating that two ratios are equal

I DO — Teacher Models

6 min

Problem: A jacket costs \$80 and is on sale for 25% off. What is the sale price?

Step 1: Find the discount amount. $25\% \text{ of } \$80 = 0.25 \times 80 = \20

Step 2: Subtract discount from original price. $\$80 - \$20 = \$60$

Answer: The sale price is \$60.

Alternative method: If 25% off, you pay 75%. $0.75 \times \$80 = \60 ✓

WE DO — Together

7 min

Problem: An electronics store sells headphones for \$120. They are on sale for 30% off. What is the sale price?

Work through together:

1. What is 30% of \$120?
2. What is the discount amount in dollars?
3. What is the final sale price?

Extension: If there is also an 8% tax on the sale price, what is the total cost?

YOU DO — Independent Practice

9 min

Solve the following. Show your work.

1. A restaurant bill is \$45. The sales tax is 8%. What is the total bill?
2. A store sells 3 notebooks for \$4.50. How much do 7 notebooks cost?
3. A bicycle originally costs \$240. It's on sale for 15% off. What is the sale price?

Practice Book Reference: Lesson 2, Questions 5–10

EXIT SLIP

5 min

A TV costs \$320. It is on sale for 15% off.

- A. What is the amount of the discount?
- B. What is the sale price?
- C. If sales tax is 7%, what is the final price you would pay?

DAY

5

Rational Numbers on the Number Line

Lesson 1.3 — Part 1 | Benchmark: MA.6.NSO.1.1, MA.6.NSO.1.2

Unit 1: Introduction — Week 1 — 45 min

BELL WORK

5 min

Question: Put these integers in order from least to greatest: 4, -2, 0, -5, 3

Answer: -5, -2, 0, 3, 4 (Negative numbers are always less than zero; the more negative, the smaller the value.)

BENCHMARK — LEARNING GOAL — LEARNING TARGETS

4 min

Benchmark

MA.6.NSO.1.1 — Extend previous understanding of numbers to define rational numbers. Plot, order, and compare rational numbers. MA.6.NSO.1.2 — Given a mathematical or real-world context, represent quantities using rational numbers.

Learning Goal

Students will plot, compare, and order rational numbers on a number line.

Target 1

I can plot rational numbers — including fractions, decimals, and negatives — on a number line.

Target 2

I can compare and order rational numbers using $<$, $>$, and $=$ symbols.

MAIN IDEA + VOCABULARY

7 min

Main Idea: Rational numbers include all integers, fractions, and decimals. To compare them, we can convert to the same form or plot them on a number line — numbers to the right are always greater.

Term	Definition
Rational number	Any number that can be written as a fraction p/q where $q \neq 0$ (includes integers, fractions, and decimals)
Integer	The set of whole numbers and their opposites: ..., -3, -2, -1, 0, 1, 2, 3, ...
Opposite	Two numbers that are the same distance from zero but on opposite sides of the number line
Absolute value	The distance a number is from zero — always positive or zero ($ -4 = 4$ and $ 4 = 4$)
Negative number	A number less than zero — located to the left of zero on the number line

I DO — Teacher Models

6 min

Problem: Plot and order these numbers: $-\frac{3}{4}$, 1.5, -1.25 , $\frac{2}{3}$

Step 1: Convert to decimals to compare. $-\frac{3}{4} = -0.75$ | $\frac{2}{3} \approx 0.67$

Step 2: Order the decimals: $-1.25 < -0.75 < 0.67 < 1.5$

Step 3: Plot on the number line — place each at the correct position between whole numbers.

Key reminder: -0.75 is GREATER than -1.25 because -0.75 is closer to zero (farther right on the number line).

WE DO — Together

7 min

Together, place these on the class number line and order them:

-0.5 | $\frac{7}{4}$ | $-\frac{3}{2}$ | 0.25

Discuss:

1. Which is farthest left? How do you know?
2. Is -0.5 greater or less than $-\frac{3}{2}$? Explain using the number line.
3. Convert $\frac{7}{4}$ to a decimal — where does it fall?

Common misconception to address: " -3 is bigger than -1 because $3 > 1$ " — use the number line to show why this is wrong.

YOU DO — Independent Practice

9 min

In your notebook, draw a number line from -3 to 3 .

1. Plot and label each number: -2 , $\frac{1}{2}$, -0.75 , 1.5 , $-1\frac{3}{4}$
2. Write them in order from least to greatest.
3. Write a comparison statement using $<$ or $>$ for: -0.75 and $-\frac{1}{2}$

Practice Book Reference: Lesson 3, Questions 1–4

EXIT SLIP

5 min

On your number line, show where -0.75 and $-\frac{1}{2}$ are located.

- A. Which is greater: -0.75 or $-\frac{1}{2}$? Write your answer using $<$ or $>$.
- B. Explain in one sentence WHY, using the words 'closer to zero' or 'farther from zero.'

Teacher Note: Students who confuse $-0.75 > -0.5$ need intervention on negative number sense before Week 2

Day 1 (absolute value).